

AGRICHAIN 360 — Reading note

This document is the **single-hub equipment annex** for the Mayuge drying hub, itemised at **UGX 12,000,000** (two hybrid solar tunnel dryers plus a smart quality-testing lab, installed).

It forms part of the hardware component of the full **UGX 45,000,000** two-hub Busoga pilot (Mayuge & Bugiri) presented in the Investment & Pilot Proposal — which additionally covers farmer onboarding, hub staffing, platform operations, compliance and contingency. Equipment specifications in this annex (ESP32 controller, moisture meter, aflatoxin testing, food-grade trays, solar power) correspond to the same platform and IoT architecture live at <https://16.192.159.6>.

Figures in this annex were costed per-hub and remain within the pilot hardware envelope (Categories A–C of the main budget).

AGRICHAIN 360™ Mayuge Agri-Hub

Complete Post-Harvest Processing Facility with IoT Quality Lab

AYuTe Africa Challenge Uganda 2026 — Season 5

TOTAL INVESTMENT

UGX 12,000,000

Two Hybrid Solar Tunnel Dryers + Full Smart Quality Testing Lab

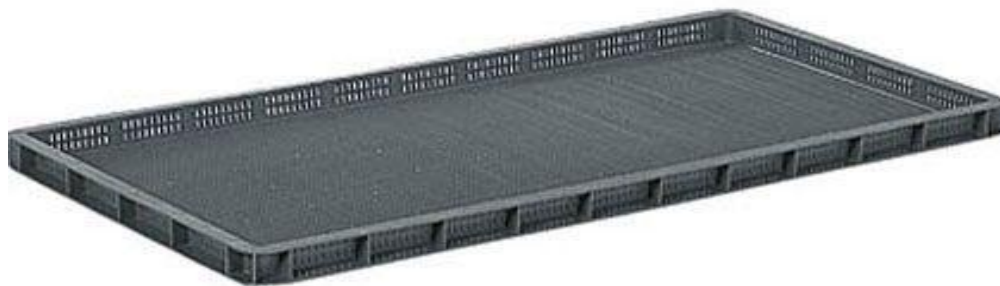
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PROJECT OVERVIEW

The **Mayuge Agri-Hub** is a centralized post-harvest processing facility featuring **two side-by-side hybrid solar tunnel dryers** — one optimized for maize and the second for coffee & cocoa. This design eliminates cross-contamination, simplifies logistics, improves security, and lowers operational costs.

The hub integrates **solar-powered drying**, **real-time IoT environmental monitoring**, and a **digital quality testing laboratory**, enabling farmers to produce export-grade crops with full traceability and compliance with EU Deforestation Regulation (EUDR).

1. Galvanized Iron (GI) Pipes & Steel Frame — UGX 2,000,000



Heavy-duty galvanized steel forms the structural backbone of both tunnels. Resistant to rust, wind, and heavy rain. Includes main frame, roof supports, cross bracing, and foundation anchors. Expected service life: 15–20 years with minimal maintenance.

2. UV Greenhouse Polythene Cover (200 micron) — UGX 1,200,000



Premium 200-micron UV-stabilized greenhouse film covers both tunnels. Creates powerful greenhouse effect, traps heat, and completely protects crops from rain, dust, insects, and birds. Long lifespan with easy replacement.

3. Food-Grade Drying Trays & Racks — UGX 800,000



Multiple perforated food-grade drying trays ensure optimal airflow. Prevents mould growth, accelerates drying time, and allows easy cleaning between batches. Essential for high-quality, uniform drying results.

4. Solar Power System (50W Panel + Battery) — UGX 900,000



Complete off-grid solar system including 50W panel, charge controller, 12V battery, and wiring. Powers exhaust fans, ESP32 controller, sensors, and GSM module. Zero electricity bills and fully sustainable.

5. DC Exhaust Fans (2 Units) — UGX 600,000



Two powerful 12V DC industrial exhaust fans continuously remove moist air from the tunnels. Dramatically improves drying speed, prevents fungal growth, and ensures uniform temperature distribution across the entire drying area.

6. ESP32 Smart IoT Controller — UGX 350,000



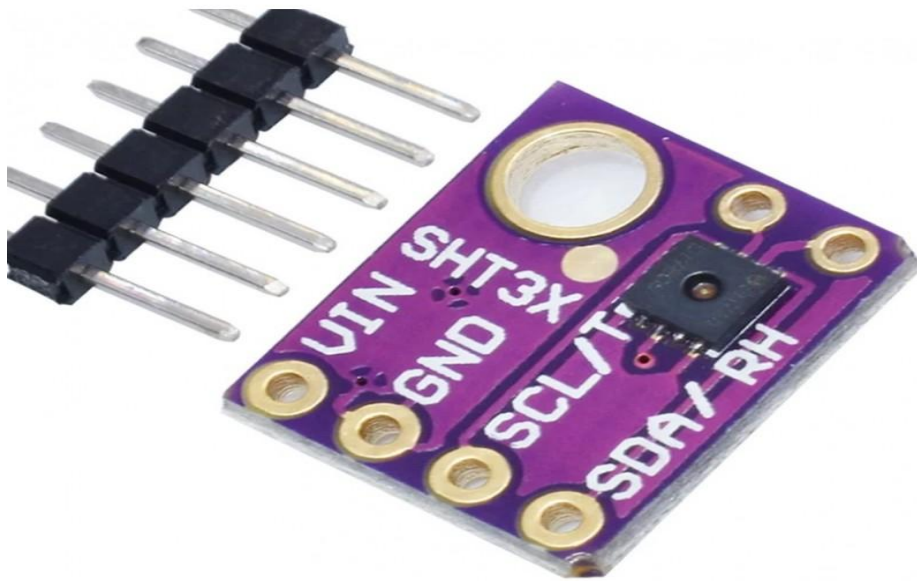
The intelligent brain of the smart dryer. Continuously reads all sensors, logs environmental data, controls fan speed, and communicates live with the AGRICHAIN 360 cloud platform via GSM for remote monitoring.

7. GSM Communication Module (SIM800L) — UGX 350,000



Reliable GSM module with MTN/Airtel SIM support. Uploads real-time sensor data to the cloud and sends instant SMS alerts to farmers and hub managers when moisture or temperature thresholds are reached.

8. Digital Grain Moisture Meter — UGX 700,000



Portable high-accuracy digital moisture meter for instant crop moisture readings. Target moisture levels: Maize 13–14%, Coffee 10–12%, Cocoa 6–8%. All readings are automatically saved to each farmer's Digital Quality Passport.

9. Aflatoxin Rapid Testing Kit — UGX 1,000,000



Professional digital aflatoxin reader with test strips. Critical for export compliance and buyer confidence. Every batch is tested before being listed on the AgriTrade marketplace. Full traceability guaranteed.

10. QR Code Label Printer + Waterproof Labels — UGX 250,000



Thermal label printer produces durable waterproof QR codes linked to each farmer's Digital Quality Passport. Contains moisture level, aflatoxin result, GPS location, harvest date, and farmer ID for complete traceability.

11. Weatherproof Electrical Housing — UGX 500,000

IP66-rated junction boxes protect all electronics from rain, dust, and insects. Includes surge protection, proper cable management, and secure mounting.

12. Tools & Installation Materials — UGX 300,000

Complete professional tool kit including drills, spanners, screwdrivers, fasteners, cable ties, measuring tape, and sealants for high-quality installation.

13. Labour & Professional Installation — UGX 750,000

Expert welding, frame fabrication, concrete anchoring, solar installation, sensor wiring, and full system testing performed by certified technicians.

COMPLETE PROCUREMENT & COST BREAKDOWN

#	Item Description	Cost (UGX)
1	Galvanized Iron (GI) Pipes & Steel Frame	2,000,000
2	UV Greenhouse Polythene Cover (200 micron)	1,200,000
3	Food-Grade Drying Trays & Racks	800,000
4	Solar Power System (50W + Battery + Controller)	900,000
5	DC Exhaust Fans (2 Units)	600,000
6	ESP32 Smart IoT Controller	350,000
7	Temperature & Humidity Sensors	300,000
8	GSM Communication Module (SIM800L)	350,000
9	Weatherproof Electrical Housing	500,000
10	Digital Grain Moisture Meter	700,000
11	Aflatoxin Rapid Testing Kit	1,000,000
12	QR Code Label Printer + Labels	250,000
13	Tools & Installation Materials	300,000
14	Labour & Professional Installation	750,000
SUBTOTAL		10,000,000
Contingency, Consumables & Training		2,000,000
GRAND TOTAL		UGX 12,000,000

REMAINING BUDGET ALLOCATION (UGX 2,000,000)

- Additional aflatoxin test strips and consumables
- Storage shelves, pallets, and batch organization system
- Personal Protective Equipment (PPE) and safety gear
- Site fencing and basic security infrastructure
- Farmer training materials and operational manuals
- Initial operating costs and contingency reserve

PROJECTED IMPACT (Year 1)

- 200+ farmers served
- 35–40% reduction in post-harvest losses
- Export-grade certification ready for EU markets
- Full blockchain traceability via QR codes
- Real-time IoT monitoring & SMS alerts

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